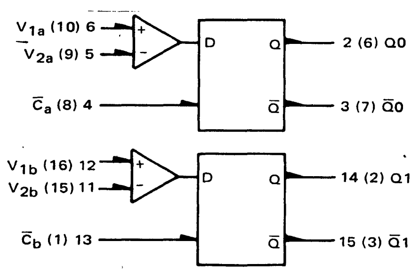
MC1650/MC1651

DUAL A/D CONVERTER



V_{CC} = +5.0 V = Pin 7,10 - (11), (14)
V_{EE} = −5.2 V = Pin 8 (12)
Gnd = Pin 1,16 (4) (5)

- P_D = 330 mW typ/pkg (No Load)
- t_{pd} = 3.5 ns typ (MC1650)
  = 3.0 ns typ (MC1651)
- Input Slew Rate = 350 V/µs (MC1650)
  = 500 V/µs (MC1651)
- Differential Input Voltage:
  5.0 V (−30°C to +85°C)
- Common Mode Range:
  −3.0 V to +2.5 V (−30°C to +85°C) (MC1651)
  −2.5 V to +3.0 V (−30°C to +85°C) (MC1650)
- Resolution: ⩽20 mV (−30°C to +85°C)
- Drives 50 Ω lines

Number at end of terminal denotes pin number for L package (Case 620).
Number in parenthesis denotes pin number for F package (Case 650).

The MC1650 and the MC1651 are very high speed comparators utilizing differential amplifier inputs to sense analog signals above or below a reference level. An output latch provides a unique sample-hold feature. The MC1650 provides high impedance Darlington inputs, while the MC1651 is a lower impedance option, with higher input slew rate and higher speed capability.

The clock inputs ($\overline{C}_a$ and $\overline{C}_b$) operate from MECL III or MECL 10,000 digital levels. When $\overline{C}_a$ is at a logic high level, Q0 will be at a logic high level provided that $V_1 > V_2$ (V_1 is more positive than V_2). $\overline{Q0}$ is the logic complement of Q0. When the clock input goes to a low logic level, the outputs are latched in their present state.

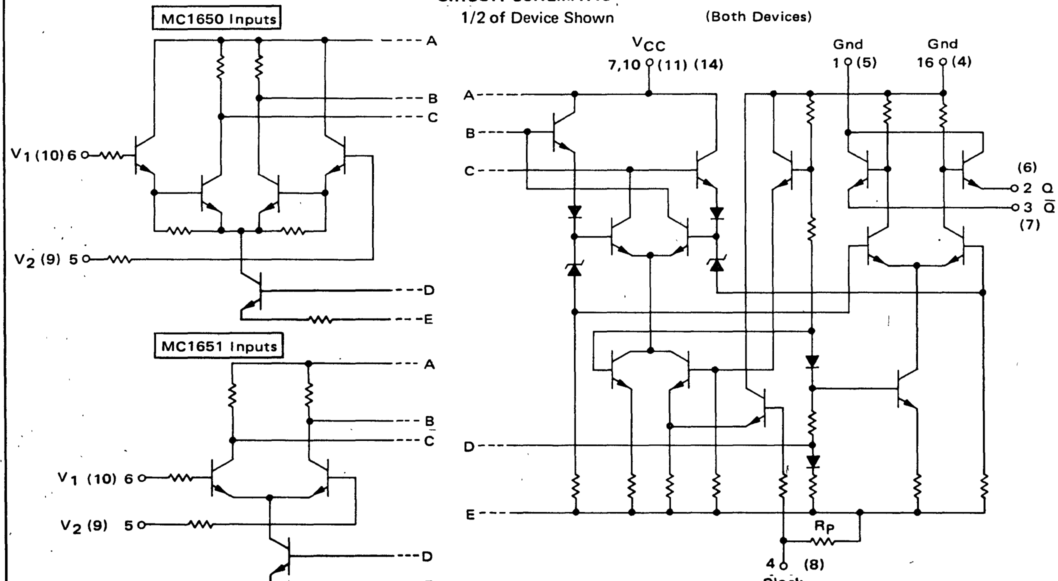
Assessment of the performance differences between the MC1650 and the MC1651 may be based upon the relative behaviors shown in Figures 4 and 7.

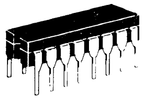
TRUTH TABLE

$\overline{C}$	V_1, V_2	$Q0_{n+1}$	$\overline{Q0}_{n+1}$
H	$V_1 > V_2$	H	L
H	$V_1 < V_2$	L	H
L	φ φ	$Q0_n$	$\overline{Q0}_n$

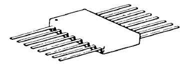
φ = Don't Care

CIRCUIT SCHEMATIC
1/2 of Device Shown (Both Devices)





L SUFFIX
CERAMIC PACKAGE
CASE 620



F SUFFIX
CERAMIC PACKAGE
CASE 650

@ Test Temperature	TEST VOLTAGE VALUES (Volts) V_{IHmax}	V_{ILmin}	V_{IHAmin}	V_{ILAmax}	V_{A1}	V_{A2}	V_{A3}	V_{A4}	V_{A5}	V_{A6}	V_{CC} (3)	V_{EE} (3)
-30°C	-0.875	-1.890	-1.180	-1.515	+0.020	-0.020	See Note (4)				+5.0	-5.2
+25°C	-0.810	-1.850	-1.095	-1.485	+0.020	-0.020					+5.0	-5.2
+85°C	-0.700	-1.830	-1.025	-1.440	+0.020	-0.020					+5.0	-5.2

ELECTRICAL CHARACTERISTICS

Characteristic	Symbol	-30°C Min	-30°C Max	+25°C Min	+25°C Max	+85°C Min	+85°C Max	Unit	TEST VOLTAGE APPLIED TO PINS LISTED BELOW V_{IHmax}	V_{ILmin}	V_{IHAmin}	V_{ILAmax}	V_{A1}	V_{A2}	V_{A3}	V_{A4}	V_{A5}	V_{A6}	Gnd
Power Supply Drain Current								mAdc											
Positive	I_{CC}	–	–	–	25*	–	–		–	4,13	–	–	6,12	–	–	–	–	–	1,5,11,16
Negative	I_E				55*				4,13	–	–	–	6,12	–	–	–	–	–	1,5,11,16
Input Current	I_{in}							μAdc											
MC1650		–	–	–	10	–	–		4	13	–	–	12	–	6	–	–	–	1,5,11,16
MC1651		–	–	–	40	–	–												
Input Leakage Current	I_R							μAdc											
MC1650		–	–	–	7.0	–	–		4	13	–	–	12	–	–	–	6	–	1,5,11,16
MC1651		–	–	–	10	–	–												
Clock Input Current	I_{inH}	–	–	–	350	–	–	μAdc	4	13	–	–	6,12	–	–	–	–	–	1,5,11,16
Logic "1" Output Voltage	V_{OH}	-1.045	-0.875	-0.960	-0.810	-0.890	-0.700	Vdc	4,13	–	–	–	6,12	–	–	–	–	–	1,5,11,16
									↓	–	–	–	–	5,11	–	–	–	–	1,6,12,16
									↓	–	–	–	–	–	6,12	5,11	–	–	1,16
									↓	–	–	–	–	–	–	–	5.11	6.12	1,16
									↓	–	–	–	–	6,12	–	–	–	–	1,5,11,16
									↓	–	–	–	5,11	–	–	–	–	–	1,6,12,16
									↓	–	–	–	–	–	5,11	6,12	–	–	1,16
									↓	–	–	–	–	–	–	–	6,12	5,11	1,16
Logic "0" Output Voltage	V_{OL}	-1.890	-1.650	-1.850	-1.620	-1.830	-1.575	Vdc	4,13	–	–	–	–	6,12	–	–	–	–	1,5,11,16
									↓	–	–	–	5,11	–	–	–	–	–	1,6,12,16
									↓	–	–	–	–	–	5,11	6,12	–	–	1,16
									↓	–	–	–	–	–	–	–	6,12	5,11	1,16
									↓	–	–	–	6,12	–	–	–	–	–	1,5,11,16
									↓	–	–	–	–	5,11	–	–	–	–	1,6,12,16
									↓	–	–	–	–	–	6,12	5,11	–	–	1,16
									↓	–	–	–	–	–	–	–	5,11	6,12	1,16
Logic "1" Threshold Voltage (2)	V_{OHA}	-1.065	–	-0.980	–	-0.910	–	Vdc	–	13	4	–	6	–	–	–	–	–	1,5,16
									–	↓	–	4	–	6	–	–	–	–	↓
									–	↓	4	–	–	6	–	–	–	–	↓
									–	↓	–	4	6	–	–	–	–	–	↓
Logic "0" Threshold Voltage (2)	V_{OLA}	–	-1.630	–	-1.600	–	-1.555	Vdc	–	13	4	–	6	–	–	–	–	–	1,5,16
									–	↓	–	4	–	6	–	–	–	–	↓
									–	↓	4	–	–	6	–	–	–	–	↓
									–	↓	–	4	6	–	–	–	–	–	↓

NOTES: (1) All data is for 1/2 MC1650 or MC1651, except data marked (*) which refers to the entire package.

(2) These tests done in order indicated. See Figure 5.

(3) Maximum Power Supply Voltages (beyond which device life may be impaired):
$|V_{EE}| + |V_{CC}| \geqslant 12$ Vdc.

(4)

All Temperatures	V_{A3}	V_{A4}	V_{A5}	V_{A6}
MC1650	+3.000	+2.980	-2.500	-2.480
MC1651	+2.500	+2.480	-3.000	-2.980

@ Test Temperature	SWITCHING TEST VOLTAGE VALUES (Volts) V_{R1}	V_{R2} V_{R3}	V_X	V_{XX}	V_{CC} (1)	V_{EE} (1)
-30°C	+2.000	See Note (4)	+1.040	+2.00	+7.00	-3.20
+25°C	+2.000		+1.110	+2.00	+7.00	-3.20
+85°C	+2.000		+1.190	+2.00	+7.00	-3.20

Characteristic	Symbol	-30°C Min	-30°C Max	+25°C Min	+25°C Max	+85°C Min	+85°C Max	Unit	Conditions (See Figures 1-3)
Switching Times Propagation Delay (50% to 50%) V-Input	t_{pd}	2.0	5.0	2.0	5.0	2.0	5.7	ns	V_{R1} to V_2, V_X to Clock, P_1 to V_1, or, V_{R2} to V_2, V_X to Clock, P_2 to V_1, or, V_{R3} to V_2, V_X to Clock, P_3 to V_1.
Clock (2)		2.0	4.7	2.0	4.7	2.0	5.2		V_{R1} to V_2, P_1 to V_1 and P_4 to Clock, or, V_{R1} to V_1, P_1 to V_2 and P_4 to Clock.
Clock Enable (3)	t_{setup}	–	–	2.5	–	–	–	ns	V_{R1} to V_2, P_1 to V_1, P_4 to Clock
Clock Aperture (3)	t_{ap}	–	–	1.5	–	–	–	ns	
Rise Time (10% to 90%)	t+	1.0	3.5	1.0	3.5	1.0	3.8	ns	V_{R1} to V_2, V_X to Clock, P_1 to V_1.
Fall Time (10% to 90%)	t-	1.0	3.0	1.0	3.0	1.0	3.3	ns	

NOTES: (1) Maximum Power Supply Voltages (beyond which device life may be impaired: $|V_{CC}| + |V_{EE}| \geqslant 12$ Vdc.

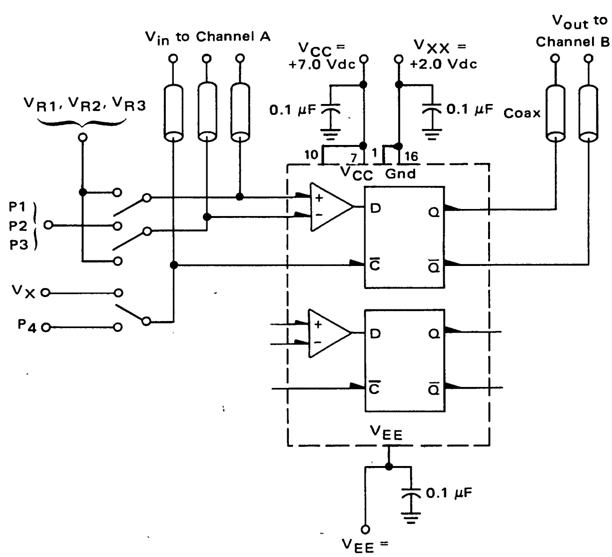
(2) Unused clock inputs may be tied to ground.

(3) See Figure 3.

(4)

All Temperatures	V_{R2}	V_{R3}
MC1650	+4.900	-0.400
MC1651	+4.400	-0.900

FIGURE 1 – SWITCHING TIME TEST CIRCUIT @ 25°C



Note: All power supply and logic levels are shown shifted 2 volts positive.

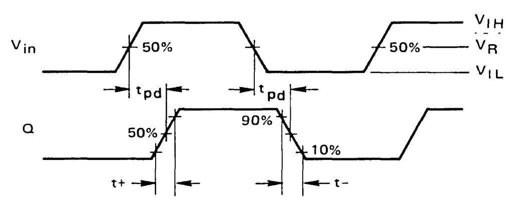
50-ohm termination to ground located in each scope channel input.

All input and output cables to the scope are equal lengths of 50-ohm coaxial cable.

FIGURE 2 – SWITCHING AND PROPAGATION WAVEFORMS @ 25°C

The pulse levels shown are used to check ac parameters
over the full common-mode range.

V – Input to Output

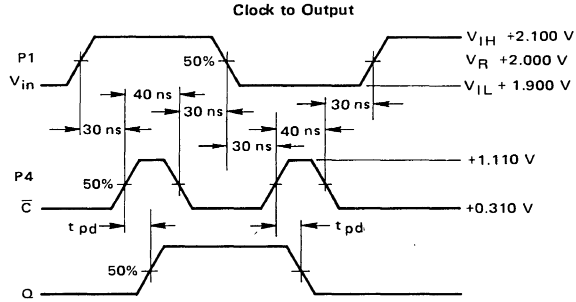


Test pulses: t_+, t_- = 1.5 ±0.2 ns (10% to 90%)
f = 5.0 MHz
50% Duty Cycle

TEST PULSE LEVELS

	P1		P2		P3	
	MC1650	MC1651	MC1650	MC1651	MC1650	MC1651
V_{IH}	+2.100 V	+2.100 V	+5.000 V	+4.500 V	-0.300 V	-0.800 V
V_R	+2.000 V	+2.000 V	+4.900 V	+4.400 V	-0.400 V	-0.900 V
V_{IL}	+1.900 V	+1.900 V	+4.800 V	+4.300 V	-0.500 V	-1.000 V

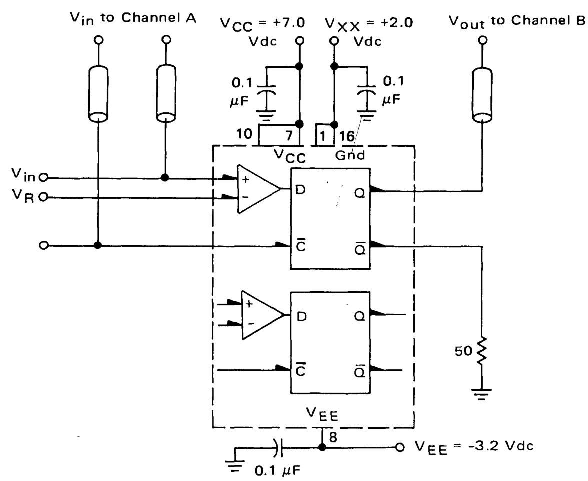
Clock to Output



P4: t_+, t_- = 1.5 ± 0.2 ns.

4

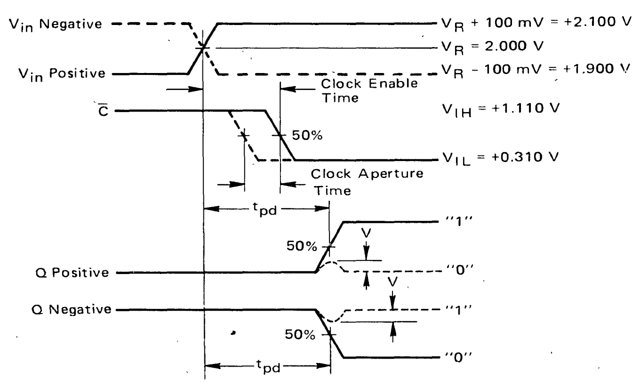
FIGURE 3 – CLOCK ENABLE AND APERTURE TIME TEST CIRCUIT AND WAVEFORMS @ 25°C



50-ohm termination to ground located in each scope channel input.

All input and output cables to the scope are equal lengths of 50-ohm coaxial cable.

Analog Signal Positive and Negative Slew Case



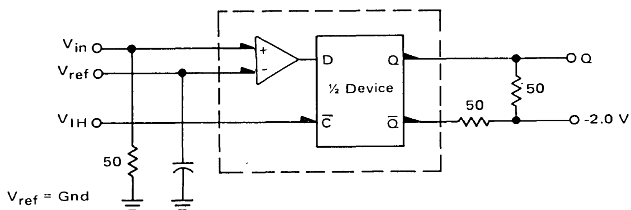
——— Clock enable time = minimum time between analog and clock signal such that output switches, and t_{pd} (analog to Q) is not degraded by more than 200 ps.

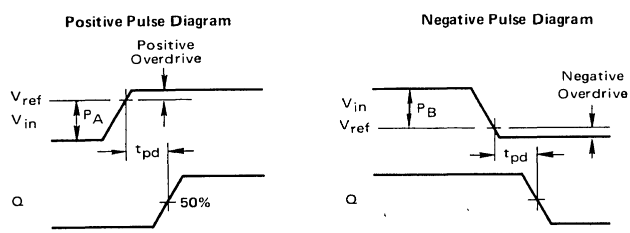
– – – – – Clock aperture time = time difference between clock enable time and time that output does not switch and V is less than 150 mV.

Note: All power supply and logic levels are shown shifted 2 volts positive.

FIGURE 4 – PROPAGATION DELAY (t_{pd}) versus INPUT PULSE AMPLITUDE AND CONSTANT OVERDRIVE

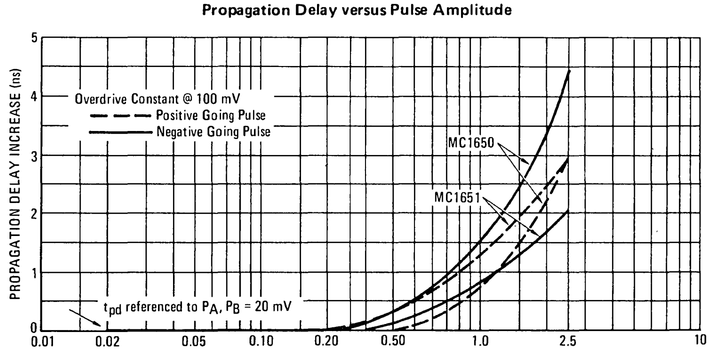
Test Circuit





Input switching time is constant at 1.5 ns (10% to 90%).

Propagation Delay versus Pulse Amplitude



(continued)

4

FIGURE 4 (continued)

Propagation Delay versus Overdrive

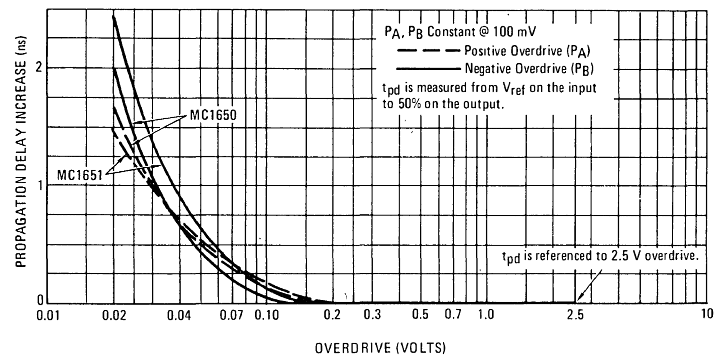


FIGURE 5 – LOGIC THRESHOLD TESTS (WAVEFORM SEQUENCE DIAGRAM)

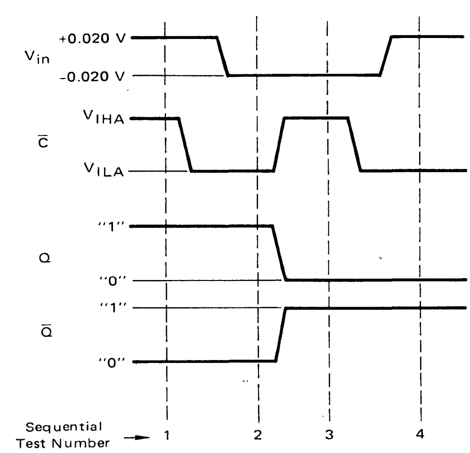
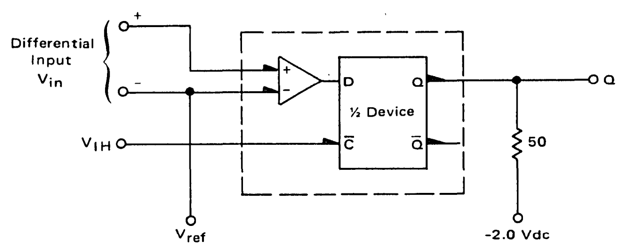


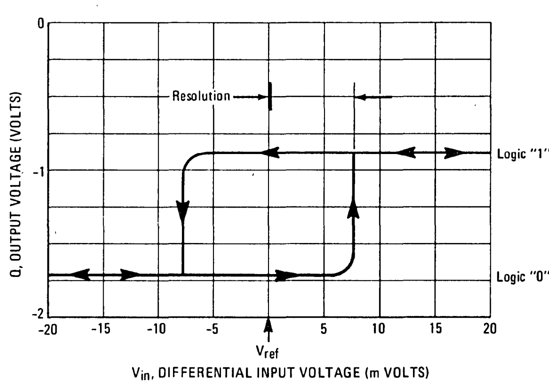
FIGURE 6 — TRANSFER CHARACTERISTICS (Q versus V_{in})

Test Configuration



$-2.5 \text{ Vdc} \leqslant V_{ref} \leqslant +2.5 \text{ Vdc}$

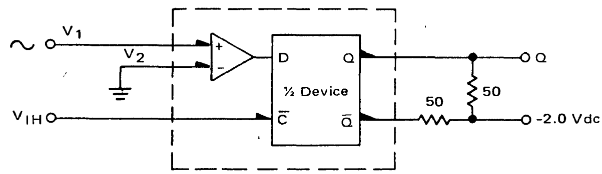
Typical Transfer Curves



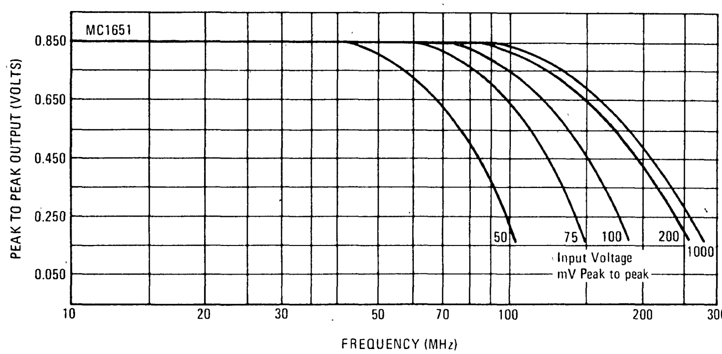
4

FIGURE 7 — OUTPUT VOLTAGE SWING versus FREQUENCY

(A) Test Circuit



(B) Typical Output Logic Swing versus Frequency



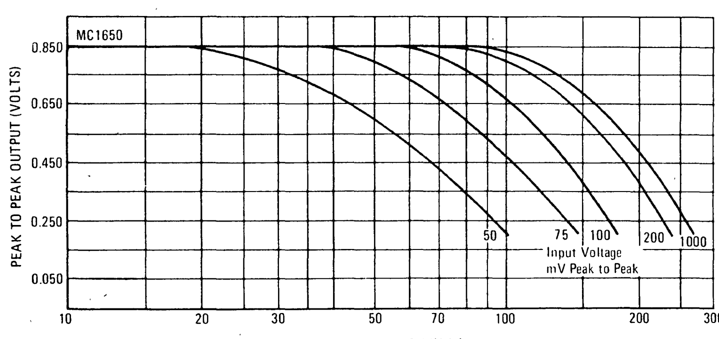
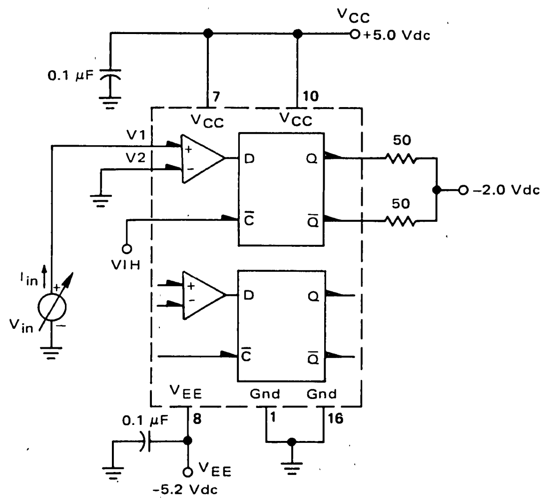
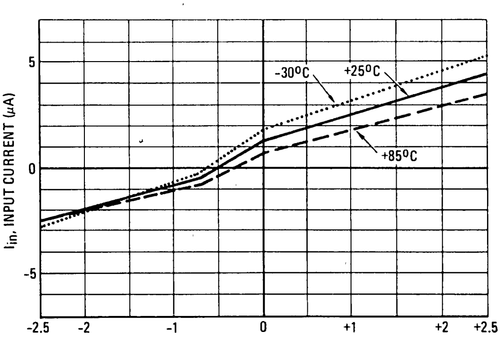


FIGURE 8 — INPUT CURRENT versus INPUT VOLTAGE

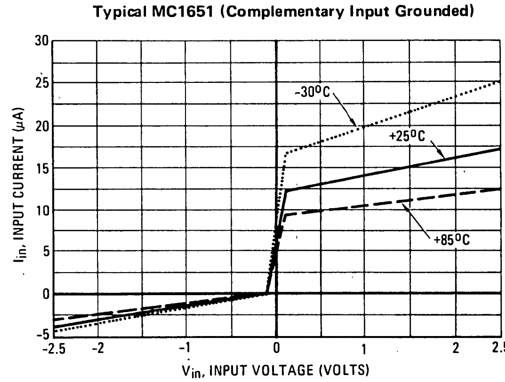
TEST CIRCUIT



Typical MC1650 (Complementary Input Grounded)



Typical MC1651 (Complementary Input Grounded)



4